3.1 칸토어 집합

정의

$C_0 = [0,1]$ 1개

$C_1 = [0,1] \setminus (\frac{1}{3}, \frac{2}{3})$ 2개

$C_2 = [0,1] \setminus (\frac{1}{3}, \frac{2}{3}) \setminus (\frac{1}{9}, \frac{2}{9}), (\frac{7}{9}, \frac{8}{9})$

then $C = \bigcap_{n=0}^{\infty} C_n$

$C = \{ 0.a_1 a_2 a_3 \cdots (3) \} \subseteq [0,1]$

$\{0, 2\}$

left, right

성질

(i) $C \neq \emptyset$ $\because$ $\forall$ endpoint $\in C$

(ii) $\text{length}(C) = 0$ $\because$ $\text{length}(\text{void}) = \frac{1}{3} + 2 \cdot (\frac{1}{9}) + 4 \cdot (\frac{1}{27}) + \cdots$

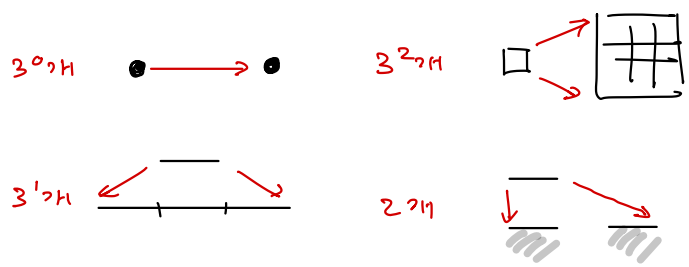
$= \frac{1}{3} \cdot \left(\frac{1}{1 - \frac{2}{3}} \right) = 1$

(iii) $\{\text{endpoints}\} \sim \mathbb{N}$ $\because$ $2 + 2 + 4 + 8 + \cdots$ $\longrightarrow$ 2^n개씩 증가

(iv) $C \sim \mathbb{R}$ $\because$ $0.\underbrace{a_1}_{2}\underbrace{a_2}_{\times 2}\underbrace{a_3}_{\times 2}\underbrace{a_4}_{\times 2} \cdots (3)$

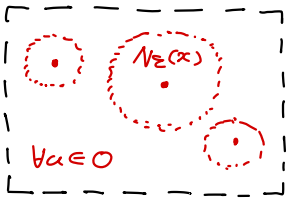
(v) C의 차원 $\because$ 3내

$= \log_3 2 \approx 0.631$



3.2 열린 집합과 닫힌 집합

열린 집합

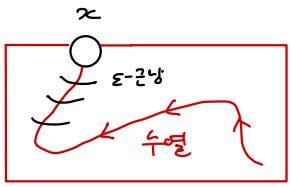


$O \subseteq \mathbb{R}$ is open : $\underbrace{\forall a \in O}_{\text{"모든 점"}}$ $\underbrace{N_\varepsilon(a)}_{\exists \varepsilon > 0}$ $\subseteq O$ (ε-근방)

(i) (합집합) $\bigcup_{\lambda \in \Lambda} \underbrace{O_\lambda}_{\text{open}}$ is open $\because$ $\underbrace{N_\varepsilon(a)}_{\varepsilon\text{-근방}} \subseteq \exists O_{\lambda'}$

(ii) (유한교집합) $\bigcap_{k=1}^{N} \underbrace{O_k}_{\text{open}}$ is open $\because$ $\underbrace{N_\varepsilon(a)}_{\varepsilon = \min(\varepsilon_1, \dots, \varepsilon_N)} \subseteq \underbrace{N_{\varepsilon_k}(a)}_{\varepsilon\text{-근방}} \subseteq \forall O_k$ (well-ordering)

극한점.



$A \setminus \{x\}$

Note.

① 극한점이 아닌 A의 원소는 고립점이다

② 극한점은 A에 포함되지 않아도 된다.

$x \in \underbrace{A'}_{\text{limit point}}$: $\underbrace{\forall N_\varepsilon(x)}_{\varepsilon\text{-근방}}$ has elements of $A \setminus \underbrace{\{x\}}_{\text{고립점}}$.

iff $\underbrace{\exists (x_n)}_{\text{수열}} \subseteq A \setminus \underbrace{\{x\}}_{\text{고립점}} \longrightarrow \underbrace{x}_{\text{극한}}$ (수열의 극한)

닫힌 집합 $F \subseteq \mathbb{R}$ is "closed" : F는 (자신의) 극한점을 모두 포함한다.

iff $\underbrace{\text{F의 원소로 이뤄진 수렴하는 수열은}}_{(x_n) \subseteq F \longrightarrow x}$ $\underbrace{\text{F 안에서 극한값을 가진다.}}_{x \in F}$

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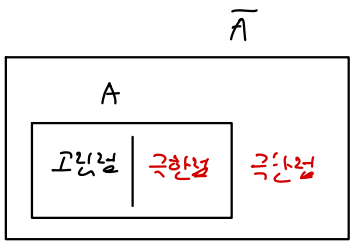
closure



- $\underbrace{\bar{A}}_{\text{closure}} = A \cup \underbrace{A'}_{\text{극한점}}$

- $x \in \bar{A}$ iff $\underbrace{\forall N_\varepsilon(x)}_{\varepsilon\text{-근방}}$ has elements of A

  iff $\underbrace{\exists (x_n)}_{\text{수열}} \subseteq A \longrightarrow \underbrace{x}_{\text{극한}}$ (수열의 극한)

$\bar{A}$는 A를 포함하는 가장 작은 닫힌 집합이다.

$\because$ $x \in \bar{\bar{A}}$ then $N_\varepsilon(x)$ has $y \in \bar{A}$ (x는 극한점에 가깝다)

$N_\varepsilon(y)$ has $z \in A$ (극한점은 A에 가깝다)

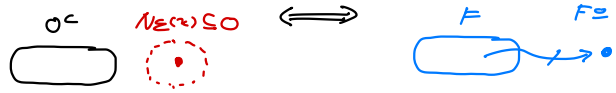
open, closed

O is open $\iff$ O^c is closed

F is closed $\iff$ F^c is open

$\because$ let $x \in O$ $\iff$ let $x \notin F$

O^c $N_\varepsilon(x) \subseteq O$ F F의 극한점 아님.



$\overline{\mathbb{Q}} = \mathbb{R}$

$\underbrace{\exists (\xi_n)}_{\text{수열}} \subseteq \mathbb{Q} \longrightarrow \underbrace{\forall r}_{\text{극한}} \in \mathbb{R}$ $\because$ $\mathbb{Q}$ is dense in $\mathbb{R}$

i.e $\underbrace{\forall a}_{\in \mathbb{R}} < \underbrace{\exists \xi}_{\in \mathbb{Q}} < \underbrace{\forall b}_{\in \mathbb{R}}$

$x \in A'$ (limit point) $\iff$ $\exists (x_n) \subseteq A \setminus \{x\} \longrightarrow x$

$x \in \bar{A}$ (closure) $\iff$ $\exists (x_n) \subseteq A \longrightarrow x$.

A is closed $\iff$ $(x_n) \subseteq A \to x$ then $x \in A$

A is bounded $\iff$ $(x_n) \subseteq A$ then $\exists (x_{n_k}) \longrightarrow x$

A is compact $\iff$ $(x_n) \subseteq A$ then $\exists (x_{n_k}) \longrightarrow x \in A$

$A = \{ \frac{1}{n} : n \in \mathbb{N} \}$ then

isolated point $= A$

limit point $= 0$

closure $= A \cup \{0\}$

$[c, d]$ is closed $\because$ $(x_n) \underbrace{\subseteq [c, d]}_{c \le x_n \le d} \longrightarrow x$ then $\underbrace{x \in [c, d]}_{c \le x \le d}$

$(c, d]$ is neither open nor closed

$\emptyset$, $\mathbb{R}$ is both open and closed

닫힌 집합

(유한 합집합) $\bigcup_{k=1}^{N} \underbrace{F_k}_{\text{closed}}$ is closed

(교집합) $\bigcap_{\lambda \in \Lambda} \underbrace{F_k}_{\text{closed}}$ is closed

3.3 콤팩트 집합.

하이네 - 보렐 정리

$K \subseteq \mathbb{R}$ is compact : $\underbrace{\forall (x_n) \subseteq K}_{K\text{의 원소로 이뤄진 수열}}$ then $\underbrace{\exists (x_{n_k}) \to x \in K}_{K\text{ 안에서 극한값을 가지는 부분수열.}}$

iff K is bounded and closed

iff K의 "임의의" open cover는 finite subcover를 가진다.

- $[c, d]$ is compact $\because$ bounded, closed
- Cantor Set is compact

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compact $\Rightarrow$ bounded & closed

$\because$ K is bounded, otherwise

$\underbrace{|x_n| > n}_{\text{not bounded}}$ then $\underbrace{\exists (x_{n_k}) \to x \in K}_{\text{compactness}}$ however (x_{n_k}) not bounded

$\because$ K is closed.

$\underbrace{(x_n) \subseteq K}_{\text{수렴하는 수열.}} \to x$ then $\underbrace{\exists (x_{n_k}) \to \overbrace{x}^{\text{수렴하는 수열의 부분수열}} \in K}_{\text{compactness}}$

축소 콤팩트 집합 성질 $\because$ $K_1 \supseteq K_2 \supseteq K_3 \cdots$

$\quad \ni x_1 \quad \ni x_2 \quad \ni x_3 \cdots$ (수열)

$\begin{cases} K_n \neq \emptyset \\ K_n \text{ compact} \end{cases}$

$K_1 \supseteq K_2 \supseteq K_3 \cdots$

then $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$

- $(x_n) \underbrace{\subseteq K_1}_{\text{compact}}$ therefore $\underbrace{\exists (x_{n_k}) \to x \in K_1}_{\text{compactness}}$
- (x_{n_k})의 꼬리 $\underbrace{\subseteq K_n}_{\text{closed}} \to \underbrace{x \in K_n}_{\text{closed}}$ $(\forall n \in \mathbb{N})$

- $\{O_\lambda\}$ is open cover of A : $A \subseteq \bigcup_{\lambda \in \Lambda} \underbrace{O_\lambda}_{\text{open}}$

- 유한개의 합집합으로 cover 가능 $\Longrightarrow$ finite subcover

---

let $O_x = \left(\frac{x}{2}, 1\right)$, $O_0 = (-\varepsilon, \varepsilon)$, $O_1 = (1-\varepsilon, 1+\varepsilon)$

$(0,1)$ open cover $\{O_x : x \in (0,1)\}$

then no finite subcover $\{O_{x_1}, \dots, O_{x_n}\}$

(b\c) $\frac{1}{2} \min\{x_1, \dots, x_n\} \notin \bigcup_{i=1}^{n} O_{x_i}$

$[0,1]$ open cover $\{O_0, O_1, O_x : x \in (0,1)\}$

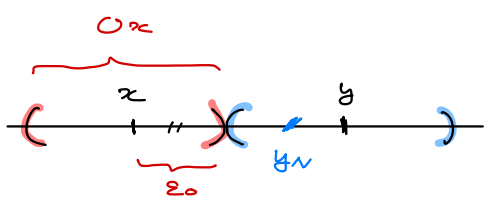
finite subcover $\{O_0, O_1, \underbrace{O_{x'}}_{\frac{x'}{2} < \varepsilon}\}$

$\ulcorner$ ∴ (③ ⇒ ②)

K is bounded. open cover $\{\underbrace{O_x}_{N_\varepsilon(x)} : x \in K\}$ has finite subcover $\{O_{x_1}, \dots, O_{x_n}\}$

K is closed otherwise suppose $(y_k) \subseteq K$ and $(y_k) \longrightarrow y \notin K$

let $O_x = N_{\frac{|x-y|}{2}}(x)$



then $\{O_x : x \in K\}$ has finite subcover $\{O_{x_1}, \dots, O_{x_n}\}$

let $\varepsilon_0 = \underbrace{\min\left\{\frac{|x_i - y|}{2}\right\}}_{\text{y에서 가장 가까운 } x_i}$ and $\underbrace{|y_{n \geq N} - y| < \varepsilon_0}_{\text{finite subcover로 넣어놓음.}}$

3.4 완전집합과 연결집합.

perfect set : closed set
without isolated points

Cantor set $C = \bigcap_{n=1}^{\infty} C_n$ is Perfect set

$\because$ ① closed $\because$ intersection of closed.

② no isolated points $\because$ $\begin{cases} x = 0.a_1a_2a_3\cdots_{(3)} \\ \text{toggle } a_n \text{ to make } x_n \neq x \end{cases}$

$\Rightarrow \exists (x_n) \subseteq \underbrace{C \setminus \{x\}}_{\text{고립점 아님.}} \longrightarrow x$

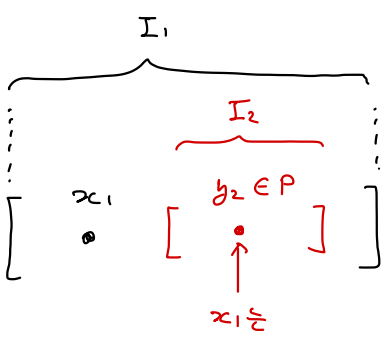
---

pefect set $(\neq \emptyset)$ is uncountable

$\because$ otherwise $P = \underbrace{\{x_1, x_2, \ldots\}}_{\text{perfect}}$

construct $K_1 \supseteq K_2 \supseteq K_3 \cdots$ $\left(\begin{array}{l} K \neq \emptyset \\ K \text{ compact} \end{array} \right)$

$\neq x_1 \quad \neq x_2$



(i) $I_{n+1} \subseteq I_n$

(ii) $x_n \notin I_{n+1}$

(iii) $K_n = I_n \cap P \neq \emptyset$ (compact)

then $\underbrace{x \in \bigcap_{n=1}^{\infty} K_n}_{x \neq x_n \ (\forall n \in \mathbb{N})} \subseteq P = \{x_1, x_2, \ldots\}$ (축도 콤팩트 집합 성질)

$A, B \subseteq \mathbb{R}$ seperated : $\bar{A} \cap B = \emptyset$

and $A \cap \bar{B} = \emptyset$

if $E = A \cup B$ then disconnected set

otherwise connected set

Note.

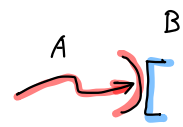
seperated $(1, 2)$ $(2, 3)$

not $(1, 2]$ $(2, 3)$

$\mathbb{Q}$ is disconnected set $\because \sqrt{2}$

$E \subseteq \mathbb{R}$ is connected $\Longleftrightarrow$ for any disjoint partition $E = A \cup B$ $(A, B \neq \emptyset)$

$\exists (a_n) \subseteq \underbrace{A}_{\text{or } B} \longrightarrow b \in \underbrace{B}_{\text{or } A}$



$E \subseteq \mathbb{R}$ is connected $\Longleftrightarrow$ E is an interval : $\forall a, b \in E$

with $a < c < b$ then $c \in E$

$\because (\Rightarrow)$

suppose $\begin{cases} E \text{ is connected} \\ c \notin E \end{cases}$

$\begin{cases} A = E \cap (-\infty, c) \neq \emptyset & (\in a) \\ B = E \cap (c, \infty) \neq \emptyset & (\in b) \end{cases}$

then $E = A \cup B$ is $\underbrace{\text{disconnected}}_{A, B \text{ are seperated}}$

$\because (\Leftarrow)$

- disjoint partition $E = A \cup B$ $(A, B \neq \emptyset)$
- $I_0 = [\overset{\in A}{a_0}, \overset{\in B}{b_0}]$

$\begin{cases} I_n\text{의 중점은 } A \text{ 또는 } B\text{에 속한다.} \\ I_n\text{을 반으로 낸다} \Rightarrow I_{n+1} = [\overset{\in A}{a_{n+1}}, \overset{\in B}{b_{n+1}}] \end{cases}$

- (축소구간정리) $x \in \bigcap_{n=0}^{\infty} I_0$

and $(a_n) \longrightarrow x$

$(b_n) \longrightarrow x$

3.5 베르 정리

- 열린 집합은 $\underbrace{\text{열린 구간}}_{\text{interval!}}$의 countable한 합집합이다.

- $\underbrace{\text{Cantor Set}}_{\text{닫힌 집합}}$ $\begin{cases} \text{is uncountable} \quad (C \sim \mathbb{R}) \\ \text{contains no intervals i.e } \operatorname{int}(C) = \emptyset \end{cases}$

  ⇒ interval을 이용한 깔끔한 특성화가 어렵다.

$$\underbrace{A}_{F_\sigma \text{ set}} = \overset{\text{countable}}{\bigcup_{n=1}^{\infty}} \underbrace{F_n}_{\text{closed}} \qquad \underbrace{B}_{G_\delta \text{ set}} = \overset{\text{countable}}{\bigcap_{n=1}^{\infty}} \underbrace{G_n}_{\text{open}}$$

Note. $\underbrace{F}_{\text{closed}} \subseteq F_\sigma$

$\underbrace{G}_{\text{open}} \subseteq G_\delta$

- B is $\underbrace{G_\delta}_{\text{open}'}$ iff B^c is $\underbrace{F_\sigma}_{\text{closed}'}$

- closed′ $\begin{cases} F_\sigma \text{의 가산 합집합} \Longrightarrow F_\sigma \\ F_\sigma \text{의 유한 교집합} \Longrightarrow F_\sigma \end{cases}$

- open′ $\begin{cases} G_\delta \text{의 가산 교집합} \Longrightarrow G_\delta \\ G_\delta \text{의 유한 교집합} \Longrightarrow G_\delta \end{cases}$

- interval is both $\underbrace{G_\delta}_{\text{open}'}$ and $\underbrace{F_\sigma}_{\text{closed}'}$ $\quad \because$ 반열린구간 $(a, b]$

$$\underbrace{\bigcap_{n=1}^{\infty} (a, b+\tfrac{1}{n})}_{G_\delta} \qquad \underbrace{\bigcup_{n=1}^{\infty} [a+\tfrac{1}{n}, b]}_{F_\sigma}$$

$\begin{cases} \mathbb{Q} \text{ is } F_\sigma \quad (\text{i.e } \mathbb{Q}^c \text{ is } G_\delta) \\ \mathbb{Q} \text{ is not } G_\delta \quad (\text{i.e } \mathbb{Q}^c \text{ is not } F_\sigma) \end{cases}$ $\quad \because \mathbb{Q} = \bigcup_{i=1}^{\infty} \underbrace{\{q_i\}}_{\text{closed}}$ with $\mathbb{Q} \sim \mathbb{N}$

$\because$ • G_1 open $\Rightarrow$ $\underbrace{\exists I_1}_{\text{closed interval}} \subseteq \underbrace{G_1}_{\text{open}}$

countable

$\bigcap_{n=1}^{\infty} \underbrace{G_n}_{\text{dense, open}} \neq \emptyset$

(actually dense)

• $\begin{cases} G_n \text{ dense} \Rightarrow I_{n-1} \cap \underbrace{G_n}_{\text{dense}} \neq \emptyset \\ G_n \text{ open} \Rightarrow \underbrace{\exists I_n}_{\text{closed interval}} \subseteq I_{n-1} \cap \underbrace{G_n}_{\text{open}} \end{cases}$

• $I_1 \supseteq I_2 \supseteq I_3 \cdots$ therefore $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ (축소구간정리)

$\mathbb{R} \neq \bigcup_{n=1}^{\infty} \underbrace{F_n}$

① closed

② open interval

Σ 불 X

$\because$ otherwise, $\mathbb{R}^c = \bigcap_{n=1}^{\infty} \underbrace{G_n}_{\text{dense, open}} \neq \emptyset$ $(G_n = F_n^c)$

if G_n not dense then $\underbrace{\exists I}_{\text{open interval}} \cap G_n = \emptyset$ therefore $\underbrace{I}_{\text{open interval}} \subseteq F_n$

$\mathbb{Q}^c$ not $\underbrace{F_\sigma}_{\text{closed}'}$

(i.e $\mathbb{Q}$ not $\underbrace{G_\delta}_{\text{open}'}$)

$\because$ we know $\mathbb{Q} = \bigcup_{i=1}^{\infty} \underbrace{\{q_i\}}$

① closed

② open interval

Σ 불 X

suppose $\mathbb{Q}^c$ is F_σ i.e $\mathbb{Q}^c = \bigcup_{k=1}^{\infty} \underbrace{K_k}_{\text{closed}}$

then K_k does not contain open interval ($\mathbb{Q}$ is dense)

$\underbrace{\mathbb{Q}}_{\text{closed}'} \cap (-\infty, 0)$ and $\underbrace{\mathbb{Q}^c}_{\text{open}'} \cap [0, \infty)$ then $A \cup B$ neither $\underbrace{F_\sigma}_{\text{closed}'}$ nor $\underbrace{G_\delta}_{\text{open}'}$

A $\quad$ B

• G dense (in $\mathbb{R}$) iff $\underbrace{\overline{G}}_{\text{근삿점}} = \mathbb{R}$

• E nowhere-dense (in $\mathbb{R}$) iff $\overline{E}$는 open interval ($\neq \emptyset$)을 Σ 불 X

iff $\overline{E}^c$ is dense $\because$ otherwise $\underbrace{\exists I}_{\text{open interval}} \cap \overline{E}^c = \emptyset$

$\begin{cases} \text{dense} : \mathbb{Q},\ \mathbb{Q}^c \\ \text{nowhere-dense} : \mathbb{N},\ \{\frac{1}{n} : n \in \mathbb{N}\},\ \text{Cantor Set} \\ \text{neither} : \text{interval} \end{cases}$

베르 정리

$$\mathbb{R} \neq \overset{\text{countable}}{\bigcup_{n=1}^{\infty}} \underbrace{E_n}_{\text{nowhere dense}}$$

otherwise $\mathbb{R} = \bigcup_{n=1}^{\infty} \underbrace{\overline{E_n}}_{\text{closure}}$

$$\mathbb{R}^c = \bigcap_{n=1}^{\infty} \underbrace{\overline{E_n}^c}_{\text{open, dense}} \neq \emptyset$$

3.6 리치보기

무한집합의 크기

- cardinality : $\text{card}(\mathbb{N}) < \text{card}(\mathbb{R})$
- 길이, 측도 : $\text{length}(\text{Cantor Set}) = 0$
- 베르 정리
  - (thin) : nowhere-dense
  - 제1범주 : (nowhere-dense)의 셀 수 있는 합집합.
  - 제2범주 : ...

$\underbrace{\text{complete}}_{\text{완비}}$ $\underbrace{\text{metric space}}_{\text{거리공간}}$

- completeness : $\forall$ Cauchy Sequence converges within that space

  i.e $\begin{cases} \mathbb{R} \text{ is complete} \\ \mathbb{Q} \text{ is not } '' \end{cases}$

- metric (d) i.e in $\mathbb{R}$, $d(x,y) = |x-y|$

Baire Category Thm.

$$\underbrace{X}_{\text{complete metric space}} \neq \overset{\text{countable.}}{\bigcup_{n=1}^{\infty}} \underbrace{E_n}_{\text{nowhere dense}}$$

- $C[0,1]$는 ($[0,1]$에서 연속인 함수들)

  완비 거리공간! $\left(d(f,g) = \sup_{x \in [0,1]} |f(x) - g(x)| \right)$

- D : 한 점에서라도 diffable 한 함수들

- $\begin{cases} D \text{ is 제1범주} \\ D^c \text{ is 제2범주} \end{cases}$

  "$\mathbb{Q}$가 $\mathbb{R}$에 차지하는 비율이 미미한 것과 같다"